Mental Health of Elite Sport Coaches and Entourage

Rosemary Purcell, MPsych, PhD[a,b,*], Joshua Frost, BSc, MSc[a,b], Vita Pilkington[a,b]

KEYWORDS

• Mental health • Wellbeing • Elite sport • Coach • Entourage • Support staff

KEY POINTS

- The mental health of elite coaches and entourage staff has been understudied, particularly in comparison to the mental health of elite athletes.
- Elite coaching and entourage staff are subject to similar sport-related stressors and mental health risk factors as elite athletes, such as performance demands, scrutiny, and travel away from home for sport.
- Available evidence, while scarce, shows that elite coaches and entourage experience depression, anxiety, and psychological distress at similar levels to elite athlete samples.
- Additional research is needed to strengthen understanding of the extent of mental ill-health and mental health needs among individuals working in elite-level sport alongside athletes.
- Based on the extant literature, mental health screening and interventions for elite athletes should be extended to the coaches and entourage staff working in the same high-performance environments.

INTRODUCTION

Research indicates that a range of sport-specific and general stressors are associated with mental health symptoms in elite athletes.[1,2] Sport-specific stressors include performance decline,[3] serious injury,[4–6] competing for selection, deselection,[7] and maladaptive perfectionism (characterized by setting unrealistic standards or seeking a high degree of control within the sport setting).[8,9] General risk indicators that may or may not be related to the sporting role include adverse life events, inadequate social support, unhelpful coping style (eg, avoidance), and sleep dysregulation.[10,11]

While elite athlete mental health has gained significant attention, the mental health of others within elite sports settings has received considerably less attention in the sports psychology literature. Although a small evidence base is developing regarding mental health among elite-level coaches, mental health among high-performance

[a] The Centre for Youth Mental Health, The University of Melbourne, Melbourne, Australia;
[b] Elite Sports and Mental Health, Orygen, 35 Poplar Road, Parkville, Victoria 3052, Australia
* Corresponding author.
E-mail address: rpurcell@unimelb.edu.au

Clin Sports Med ■ (2023) ■–■
https://doi.org/10.1016/j.csm.2023.06.005 sportsmed.theclinics.com
0278-5919/23/© 2023 Elsevier Inc. All rights reserved.

Downloaded for Anonymous User (n/a) at The Ohio State University from ClinicalKey.com by Elsevier on October 20, 2023. For personal use only. No other uses without permission. Copyright ©2023. Elsevier Inc. All rights reserved.

support staff (HPSS; referred to throughout this article as "entourage") has been neglected. Entourage members include physiotherapists, strength and conditioning coaches, athletic trainers, and other allied health practitioners, such as nutritionists, sport psychologists, sports medicine physicians, player development managers, and athlete wellbeing advisors. Coaches and sports entourage staff operate within the broader "ecology" of elite sporting environments[12] and have critical roles to play in supporting the mental health of elite athletes.

Ecological systems can be used to conceptualize the relationship between the aspects or experiences of an individual and the broader socio-cultural contexts in which they work or operate.[13] Coaches and entourage (along with teammates and parents/family where appropriate) function as the core "microsystem" to support and enable elite athletes. Coaches and entourage interact with athletes in the daily training and competitive environments and therefore are often best placed (along with family or loved ones) to observe changes in an athlete's behavior, mental state, or demeanour (or in the case of some health staff, non-observable signs such as changes in muscle/body tension). Given the leadership and support roles that they occupy (in terms of supporting athletes' health, functioning, and performance), coaches and entourage can be critical to the early detection of mental ill-health and encouraging and supporting athletes to seek professional help.[12,14]

In comparison to the literature of elite athlete mental health, there has been a paucity of attention to the mental health of elite coaches and entourage despite them operating in the same high-performance environments and being subject to high expectations regarding individual or team performance outcomes. We speculate that elite coaches may be more susceptible to mental health challenges than their elite athletes because they encounter additional potential stressors in their roles, such as organisational responsibilities (including reporting to executives and boards); acting as the "public face" or spokesperson for their sport/team; educating, mentoring, and motivating their staff and athletes; long working hours; insecure employment; feeling undervalued in their role; and intense media scrutiny.[15–19] These are in addition to their key roles as technical and strategic experts. Finally, coaches and entourage also may have to contend with various personal stressors (often exacerbated by long work hours or role stress and scrutiny), such as social isolation and relationship difficulties.[17,20–23] Given the confluence of factors potentially impacting elite coaches, the lack of research consideration directed toward their mental health is surprising.

Prevalence and Risk Factors for Mental Health Symptoms in Elite Coaches

Several reviews have explored stressors and psychological health among elite coaches; however, most have focused on the constructs of wellbeing, coping, and burnout,[24–28] rather than mental health symptoms or disorders. The handful of studies that have considered common forms of mental ill-health—such as anxiety, depression, psychological distress, and alcohol use—are reviewed in the following sections. We provide the caveat that most participants in these research studies (over 70%) are men, rather than women. The overrepresentation of men in the literature is consistent with the broader gender imbalance observed in sports psychology literature[29] but also reflects the greater representation of men among elite coaching ranks.[30] Future research examining the mental health experiences of elite women coaches is a priority.

Anxiety and depression

A handful of studies have examined the prevalence of anxiety and depressive symptoms in elite coaching cohorts, with mixed results. Kegelaers and colleagues[22]

Downloaded for Anonymous User (n/a) at The Ohio State University from ClinicalKey.com by Elsevier on October 20, 2023. For personal use only. No other uses without permission. Copyright ©2023. Elsevier Inc. All rights reserved.

reported that 39.5% of a large sample of Dutch and Flemish coaches (n = 119) reported anxiety and depression symptoms (assessed using the 12-item General Health Questionnaire), whereas in a smaller sample of New Zealand coaches (n = 69), Kim and colleagues[31] found that only 14.1% met the threshold for at least moderate depressive symptoms (assessed using the Center for Epidemiologic Studies Depression Scale-Revised). Among coaches seeking treatment for mental ill-health (n = 34), Åkesdotter and colleagues[32] found that the vast majority (93%) were experiencing clinical levels of anxiety, while 28% met diagnostic criteria for major depression.

Based on available research, known correlates of anxiety and depression symptoms among elite coaches include goal incongruence, poor coping effectiveness, contemplating retirement, family history of mood disorder, and increased frequency of daily hassles.[31,33] Burnout is also associated with mental ill-health among elite coaches and was found in two studies to be the strongest predictor of anxiety and depression,[34,35] although we note that burnout can be both a predictor and a consequence of mental ill-health.

General psychological distress

Rates of "high" psychological distress (eg, broadly defined as feelings of emotional pressure, anxiety, and/or reduced ability to cope with life) range from 10.3% to 19.3% in elite coaches,[22,23,34–36] although we note that different measures of psychological distress have been used across studies, which limits comparisons. Protective factors associated with reduced psychological distress among elite coaches include satisfaction with life balance, satisfaction with social support, and older age,[23] whereas one longitudinal study indicated that psychological distress was found to peak during the competitive mid-season in comparison to preseason.[35]

Substance use

Two studies have examined risky alcohol consumption in elite coaches using the same measurement tool (the Alcohol Use Disorder Identification Test-Concise); however, different cut-off scores were applied, resulting in discrepant results. The reported rate of risky alcohol consumption was 19.3% in the study by Kegelaers and colleagues[22] versus 48.3% in the study by Pilkington and colleagues.[23] The study by Åkesdotter and colleagues using a sample of mental health help-seeking coaches found that 17% met the diagnostic criteria for a substance-use disorder.[32] Most other research related to substance use in elite sport has largely focused on athletes' substance use, including reported alcohol consumption and doping behaviors and attitudes.

Other forms of mental ill-health

One study found that risk of eating disorder in a small cohort of National Collegiate Athletics Association Division I coaches (n = 21) was 38.1%.[37] Although the study by Åkesdotter and colleagues[32] also assessed eating disorder symptomatology among elite coaches (who were seeking outpatient mental health support), the number of coaches meeting eating disorder criteria was too small to report. However, Åkesdotter's study did find a high prevalence of stress and adjustment disorders among elite coaches, with 72% of coaches presenting to the service meeting criteria for stress-related disorders (including acute stress reactions, adjustment disorder, post-traumatic stress, other reactions to severe stress), relative to only 25% of elite athletes meeting criteria for stress-related disorders at the same service.

Sleep disturbance

While not a diagnosable mental disorder, sleep disturbance is recognized as both an antecedent and a consequence of mental ill-health.[38] Two studies reported similar

Downloaded for Anonymous User (n/a) at The Ohio State University from ClinicalKey.com by Elsevier on October 20, 2023. For personal use only. No other uses without permission. Copyright ©2023. Elsevier Inc. All rights reserved.

prevalence rates of sleep disturbance in elite coaches (23.4% and 25.2%, respectively).[22,23] Several studies have reported an association between sleep disturbance and burnout.[39–41]

Burnout

Burnout is also not a mental health disorder but is one of the most frequently studied health outcomes in elite coaches and has been shown to act as a key correlate of mental ill-health in a number of studies. According to the World Health Organization, burnout refers to chronic workplace stress characterized by; 3 domains: emotional and physical exhaustion increased feelings of negativity or cynicism or mental distance from one's role: and a decrease in professional efficacy or accomplishment.

Studies that have investigated burnout in elite coaches report heterogeneous results, ranging from low,[42–47] through moderate,[42,46,48,49] and high[49] levels of burnout. In a large cohort of Finnish elite-level coaches (n = 499), 22% met criteria for mild burnout symptoms and 2% for severe symptoms.[44]- Correlates of high exhaustion in coaches at risk of burnout include negative perceptions of the sport organization's management and leadership, high workload, and performance outcomes (athlete win or loss records).[39]

Summary

While a growing interest in elite coach mental health is apparent, mental health among elite coaching staff remains poorly characterized, particularly relative to the elite athletes they support and mentor. Available data show significant variation in the reported rates of mental health symptoms among elite-level coaches, reflecting differences in sampling and measurement across studies. These methodological inconsistencies can be remediated by future studies consistently employing validated measures to assess mental health and wellbeing outcomes in sporting contexts (see Gouttebarge et al, [50]). To date, entourage staff have rarely been included in mental health literature.

AN EMPIRICAL STUDY OF MENTAL HEALTH IN ELITE COACHES AND SPORTS ENTOURAGE STAFF

The burgeoning research attention to the mental health of elite coaches has not been matched by consideration of other staff working within high-performance sport, despite the fact that elite coaches and entourage share many role-related stressors, including high workloads, competitive performance expectations, role insecurity, travel away from home, and social isolation.[51] To our knowledge, no empirical study has examined the prevalence of mental ill-health in HPSS and directly compared this to elite coaches, prior to our 2022 study.[23]

We surveyed a cohort of coaches (n = 78; mean age 46 years, 24% female) and HPSS (n = 174: mean age 40 years, 57% female) working in Australia's high-performance sports system to understand the prevalence and correlates of mental health symptoms and to explore similarities and differences in the mental health profiles between groups. The survey was conducted between March and May of 2020 and, therefore, coincided with the COVID-19 pandemic and associated lockdowns and interruptions to competitive sport. Key outcome measures included psychological distress, "caseness" (defined as symptoms of depression and anxiety at a level that would usually warrant treatment by a health professional), alcohol consumption, and sleep dysregulation. In addition to assessing the prevalence of mental health symptoms and wellbeing outcomes, we examined a range of possible correlates of

Downloaded for Anonymous User (n/a) at The Ohio State University from ClinicalKey.com by Elsevier on October 20, 2023. For personal use only. No other uses without permission. Copyright ©2023. Elsevier Inc. All rights reserved.

these mental health and wellbeing outcomes. Possible correlates included demographic variables (eg, age, gender, relationship status, duration working in high-performance sport), adverse life events, overall satisfaction with life, satisfaction with social support, satisfaction with life balance, quality of life, and level of concern about COVID-19.

As shown in **Table 1**, the rates of mental health symptoms were consistent between elite level coaches and entourage/HPSS, with no statistically significant differences between groups across all outcome measures.

In addition to reporting similar levels of mental health symptoms, coaches and entourage also endorsed similar adverse life events, including almost half who reported feeling undervalued or underpaid (across the lifetime). The most robust correlates of coach and entourage mental health outcomes were satisfaction with life balance and satisfaction with social support. Life balance emerged as a widespread issue among the sample, with just over half the sample reporting they felt satisfied with their life balance at the time of the survey and others reporting the need for greater life balance.

The only demographic variable associated with a mental health outcome (in this instance, psychological distress) was age, with younger coaches and entourage reporting elevated levels of distress. Interestingly, no gender differences were found in this survey, which contrasts with the findings among elite athlete samples, where elevated rates of mental health symptoms are observed in female athletes.[52] This suggests a differential role of gender between athletes and members of the daily training environment that warrants further research attention.

The results of this study also demonstrate the relative consistency between the rates of mental health symptoms in coaches and entourage and those of the elite athletes with whom they work alongside. Reported rates were similar across probable caseness (43.6% of coaches, 40.1% of entourage, 35% of athletes[10]), psychological distress (10.3% of coaches, 15.5% of entourage, 17.7% of athletes[10]), and sleep disturbance (23.4% of coaches, 15.1% of support staff, 16.0% of athletes[53]), although risky drinking was reported at higher rates among coaches (48.1%) and entourage (39.0%), relative to published athlete samples (25.8%[54]). This reinforces the notion that stressors encountered within elite sports settings contribute to mental health difficulties among those regularly operating within these systems.[12]

Table 1
Prevalence of reported mental health symptoms among coaches and sports entourage staff from Pilkington et al[23]

Measure	Coaches (%)	Entourage (%)
Probable caseness	43.6	40.1
High to very high psychological distress	10.3	15.5
Moderate to severe sleep disturbance	23.4	15.1
Risky alcohol consumption	48.1	39.0

"Probable caseness" refers to the proportion of participants reporting mental health symptoms at levels that usually warrant psychological support and was measured by the General Health Questionnaire-28 item version (GHQ-28). Scores of 5 or higher (with binary coding) indicated probable caseness. Psychological distress was measured by the Kessler-10 item version (K-10). Scores between 22 and 50 indicated "high to very high" psychological distress. Sleep disturbance was assessed using the Athlete Sleep Screening Questionnaire (ASSQ). Scores between 8 and 17 indicated moderate to severe sleep disturbance. Alcohol consumption was assessed using the Alcohol Use Disorders Identification Test-Condensed version (AUDIT-C). Scores of 4 or more for female staff and 5 or more for male staff indicated risky alcohol consumption.

Downloaded for Anonymous User (n/a) at The Ohio State University from ClinicalKey.com by Elsevier on October 20, 2023. For personal use only. No other uses without permission. Copyright ©2023. Elsevier Inc. All rights reserved.

CASE EXAMPLE: TOM

Tom was a 37-year-old sports medicine physician working for a major national professional sport. He was married with 2 elementary-school-aged children. His role required significant travel, nationally and occasionally internationally, and he was often away from home for more than 5 months of the year. In addition to his clinical role, Tom held an academic appointment, supervised junior sports medicine practitioners, and maintained a small private practice.

Tom described his sports medicine career as "busy but for the most part completely fulfilling." He thrived in the high-performance environment and enjoyed the deep connections and comradery he established with his colleagues and the national team players with whom he traveled extensively. He was mindful of maintaining boundaries in his role and so rarely socialized outside the team environment (eg, only celebrating wins in the locker room after games) or consumed alcohol, which, in his opinion, was a highly problematic aspect of the sporting culture. Tom exercised regularly and communicated daily with his wife and children (and his aging parents) when traveling. He described feeling guilty about missing family events, such as his children's birthdays and wedding anniversaries, due to travel, but felt he was contributing to his family by providing an income that supported a comfortable lifestyle for his wife and children.

According to Tom, engaging in robust discussions with team management, coaches, and players about the fitness of players to compete was a regular element of his role. He took his ethical responsibilities to protect player health seriously and was relaxed that he occasionally had to make unpopular calls about players' availability due to injury and/or incomplete recovery. Tom felt he had earned the trust and respect of his colleagues after more than 6 years working with the national team. This shifted, however, when Tom experienced an unsettling event in which he was verbally abused on field by a player forcefully resisting leaving play for a mandatory assessment following a heavy collision. The abuse was brief, but personal and intense, and televised. Tom felt shocked and humiliated by the encounter, which was exacerbated by the player's subsequent refusal to apologize, believing that Tom cost the team the game by removing him from play at a crucial stage. Tom complained about the abuse to the coach and team manager (citing workplace safety rules), but not only were his concerns dismissed, the coach and manager sided with the player's views that Tom's actions probably cost them the game. Tom was deeply troubled that his decision-making and ethical responsibilities were questioned. He described this being the start of "the sleepness nights" and second guessing his value to the team. He acknowledged that he ruminated about the encounter long after the player, coach, and manager had moved on from it.

Tom continued to travel with the team but felt somewhat "disconnected" from the players and coaching staff. He started to "lose the love" for the sport and increasingly spent time in his hotel room or local cafes. He neglected exercising and gained a few pounds for the first time. Tom began to take sleeping tablets to fall asleep but woke early each morning (around 4 AM) unable to get back to sleep. His wife complained about his being "even more remote than usual," which triggered an argument in which Tom was blindsided by his wife's resentment at having to "functionally raise the kids alone!", when, from his perspective, he was "busting a gut" to financially establish his family's future. That night, Tom described sitting in the backyard, uncharacteristically drinking whiskey, until the early hours of the morning "wondering whether it was all worth it anymore." When his wife came down to find him, she was shocked to find Tom silently crying. The next morning, with his wife's and dad's encouragement, Tom spoke to a sport psychiatrist he knew outside of his sport. Tom knew that he

Downloaded for Anonymous User (n/a) at The Ohio State University from ClinicalKey.com by Elsevier on October 20, 2023. For personal use only. No other uses without permission. Copyright ©2023. Elsevier Inc. All rights reserved.

was struggling and felt that he was probably burned out, but he was surprised that he scored in the "severe" range of the depression measure that the psychiatrist administered. Tom subsequently spoke confidentially to the team manager about needing to take some time away for health reasons and was supported to do so.

RISK AND PROTECTIVE FACTORS FOR MENTAL HEALTH IN ELITE COACHES AND ENTOURAGE

Workload and Recovery

Given the myriad of responsibilities for which coaches and HPSS are accountable (including managing athlete performance, condition, and leading tactical strategies), individuals within these roles commonly report highly demanding workloads.[51] Excessive workloads can emerge from role overload,[41] role conflict or ambiguity,[55] or attempts to compensate for a lack of role-based experience or resources.[39] Research indicates that as workloads intensify, coaches and entourage may be susceptible to lower wellbeing, poor mental health, and symptoms of burnout.[43,56]

Despite the demanding workloads, there is evidence to suggest that recovery processes (eg, psychological, physical, social) may play a leading role in shaping the mental health of high-performance entourages.[57,58] For example, Kellmann and colleagues found in a cohort of professional Australian football coaches that although stress levels remained relatively stable over the course of a season, recovery levels decreased throughout the competition.[58] This finding indicates that members of the sporting entourage should be provided with opportunities to psychologically detach and physically recuperate from their roles, as a lack of recovery time may contribute toward the onset of burnout.[40,41,43]

Psychological Skills and Coping Strategies

A range of individual-level characteristics have been found to influence the mental health of high-performance coaches and entourage. Several psychological characteristics have been found to protect entourage members from experiencing mental ill-health, including traits such as hardiness and resilience, which have been found to negatively predict symptoms associated with burnout, anxiety, and depression.[22,59] Conversely, entourage who possess low emotional intelligence or traits associated with maladaptive perfectionism may be at greater risk of experiencing mental ill-health.[40,41,60] Given that psychological skills can be shaped and nurtured over time, it is critical that coaches and support staff are offered opportunities to expand upon their repertoire of skills. When provided with training, mindfulness-based interventions have shown promise in significantly reducing negative affective states and symptoms of anxiety.[61] Hägglund and colleagues, for instance, found that mindful self-reflection cultivated greater levels of self-awareness in elite-level coaches.[62] Participants also reported that self-awareness helped to manage daily stressors more effectively. Because current evidence suggests that effective coping can enhance hedonic and eudaimonic wellbeing,[21,42] while ineffective coping may lead to the development of emotional exhaustion or the proliferation of symptoms associated with anxiety,[33,42] it is imperative that elite-level coaches and HPSS possess appropriate psychological tools that can be employed to manage the plethora of performance, organizational, and personal stressors that persist.

Social Support

The presence of robust social support is a key protective factor in mitigating poor mental health among elite coaches and entourage. An entourage member's support network may comprise both formal (eg, mentors) and informal (eg, friends and family)

Downloaded for Anonymous User (n/a) at The Ohio State University from ClinicalKey.com by Elsevier on October 20, 2023. For personal use only. No other uses without permission. Copyright ©2023. Elsevier Inc. All rights reserved.

sources of social support.[51] Satisfaction with one's social support has been negatively associated with a range of mental health outcomes, including symptoms of anxiety, depression, psychological distress, sleep disturbance, and burnout.[23,33,41,46,59] Considering the link between insufficient social support and the onset or development of mental ill-health,[46,59] it is important for sporting organizations to be mindful of the social supports available to their high-performance employees. This support is invaluable in safeguarding the mental health of a coach or staff member not only throughout their tenure but also after employment or competition.[63] For example, Bentzen and colleagues[64] and Kenttä and colleagues[65] reported the pivotal role social support plays in preserving the mental wellbeing of elite-level coaches who have been dismissed from their role. These supports have been found to assist coaches with reflecting, recovering, and experiencing positive emotions during this transitional phase. Given the value of maintaining robust social networks both during and outside of one's involvement in elite sport, quality social supports should be considered sacrosanct in safeguarding the mental health of high-performance coaches and support staff.

Organizational Support

Organizations and governing bodies are integral to supporting the mental health of elite-level coaches and entourage. Given organizational demands have been found to significantly predict symptoms of anxiety and depression in elite-level coaches (unlike performance and personal stressors),[22] organizations should be cognisant of the role they play in alleviating or exacerbating symptoms of mental ill-health. Research has shown that the perceived satisfaction from upper management,[48] autonomy-supportive environments,[56] and sufficient financial resources[49] can operate as protective factors for elite-level coaches. Hill and colleagues also found that organizations which promote together and belonginess, cultivate a challenging yet supportive environment, and possess a clear organizational vision or philosophy are more likely to protect the mental health and wellbeing of high-performance staff.[51] Organizations should subsequently seek to define and establish these core values, to ensure role-based expectations are realistic and understood and that coaches and support staff feel part of a cohesive and supportive working environment which fosters psychological protection via relatedness.[66]

SUPPORTING ELITE-LEVEL COACH AND ENTOURAGE MENTAL HEALTH

Mrazek and Haggerty proposed that mental health support can be classified into three broad stages of intervention—prevention, treatment, and maintenance.[67] Considering these foci, Frost and colleagues explored several potential measures that could be implemented when developing an early intervention framework for elite-level coaches.[68] It is proposed that mental health screening and monitoring, mental health literacy, psychological safety, and pathways to mental health supports warrant consideration. These measures should be equally applied to elite coaches and support staff as they encounter similar stressors (eg, high workloads, job insecurity, lack of role clarity)[23,63] and report similar rates of mental health symptoms.

Despite the utility and applicability of these strategies, at present, a greater body of evidence is needed to design and adapt these measures to specifically accommodate elite-level entourages. At the Canadian intercollegiate level, Sullivan and colleagues found no significant differences in mental health literacy between coaches and athletic therapists.[69] Results indicated, however, that female professionals possessed significantly higher levels of mental health literacy than their male counterparts and that a

Downloaded for Anonymous User (n/a) at The Ohio State University from ClinicalKey.com by Elsevier on October 20, 2023. For personal use only. No other uses without permission. Copyright ©2023. Elsevier Inc. All rights reserved.

significant negative correlation was identified between mental health literacy and total coaching experience. Future research should replicate similar procedures to evaluate mental health literacy levels and specific demographic and background-related variations among individuals operating within elite sport. This evidence may help inform organizations with tailoring mental health literacy programs to meet the specific needs of coaches and support staff.

Finally, psychological safety is increasingly recognized as critical to creating mentally healthy elite sporting environments.[70] Psychometric tools such as the Sport Psychological Safety Inventory could be used to measure psychological safety in high-performance environments,[71] including among elite-level coaches and HPSS with whom the tool was validated. This information can potentially shed light upon a coach's or support staff's sense of security in sharing sensitive information or raising issues with management,[72] subsequently providing partial insights into the degree to which these individuals feel supported by their organization. Further research is ultimately needed, however, to evaluate the perceived support that high-performance coaches and support staff receive with regard to their mental health.

SUMMARY

Elite sporting coaches and entourage operate in the same high-performance environments as elite athletes and, therefore, encounter similar risk factors for mental health as the athletes that they coach, train, and support. Despite this, there has been less research attention to the mental health of elite coaches and entourage, beyond studies examining burnout and wellbeing. A greater focus on understanding the prevalence of a range of mental health symptoms in these cohorts and the factors that contribute to vulnerability or exert a protective influence on mental health is necessary to support best-practice prevention and early intervention strategies.

CLINICS CARE POINTS

- A range of dynamic risk and protective factors have been identified that can be addressed to improve the mental health of elite coaches and entourage.
- Mindfulness based interventions and psychological skills training show promise for supporting mental welbeing among elite coaches.

DISCLOSURE

The authors have nothing to disclose.

REFERENCES

1. Reardon CL, Hainline B, Aron CM, et al. Mental health in elite athletes: International Olympic Committee consensus statement (2019). Br J Sports Med 2019;53(11):667–99.
2. Küttel A, Larsen CH. Risk and protective factors for mental health in elite athletes: a scoping review. Int Rev Sport Exerc Psychol 2020;13(1):231–65.
3. Hammond T, Gialloreto C, Kubas H, et al. The prevalence of failure-based depression among elite athletes. Clin J Sport Med 2013;23(4):273–7.
4. Gulliver A, Griffiths KM, Mackinnon A, et al. The mental health of Australian elite athletes. J Sci Med Sport 2015;18(3):255–61.

Downloaded for Anonymous User (n/a) at The Ohio State University from ClinicalKey.com by Elsevier on October 20, 2023. For personal use only. No other uses without permission. Copyright ©2023. Elsevier Inc. All rights reserved.

5. Rice SM, Gwyther K, Santesteban-Echarri O, et al. Determinants of anxiety in elite athletes: a systematic review and meta-analysis. Br J Sports Med 2019;53(11):722–30.
6. Peluso MAM, Andrade LHSGd. Physical activity and mental health: the association between exercise and mood. Clinics 2005;60(1):61–70.
7. Blakelock DJ, Chen MA, Prescott T. Psychological distress in elite adolescent soccer players following deselection. J Clin Sport Psychol 2016;10(1):59–77.
8. Jensen SN, Ivarsson A, Fallby J, et al. Depression in Danish and Swedish elite football players and its relation to perfectionism and anxiety. Psychol Sport Exerc 2018;36:147–55.
9. Jordana A, Ramis Y, Chamorro JL, et al. Ready for Failure? Irrational Beliefs, Perfectionism and Mental Health in Male Soccer Academy Players. J Ration Emot Cogn Behav Ther 2022;41:1–24.
10. Purcell R, Rice S, Butterworth M, et al. Rates and correlates of mental health symptoms in currently competing elite athletes from the Australian National high-performance sports system. Sports Med 2020;50(9):1683–94.
11. Gouttebarge V, Castaldelli-Maia JM, Gorczynski P, et al. Occurrence of mental health symptoms and disorders in current and former elite athletes: a systematic review and meta-analysis. Br J Sports Med 2019;53(11):700–6.
12. Purcell R, Gwyther K, Rice SM. Mental health in elite athletes: increased awareness requires an early intervention framework to respond to athlete needs. Sports medicine-open 2019;5(1):1–8.
13. Bronfenbrenner U. Ecological systems theory. London, England: Jessica Kingsley Publishers; 1992.
14. Chang C, Putukian M, Aerni G, et al. Mental health issues and psychological factors in athletes: detection, management, effect on performance and prevention: American Medical Society for Sports Medicine Position Statement—Executive Summary. Br J Sports Med 2020;54(4):216–20.
15. Didymus FF. Olympic and international level sports coaches' experiences of stressors, appraisals, and coping. Qualitative Research in Sport, Exercise and Health 2017;9(2):214–32.
16. Knights S, Ruddock-Hudson M. Experiences of occupational stress and social support in Australian Football League senior coaches. Int J Sports Sci Coach 2016;11(2):162–71.
17. Thelwell RC, Weston NJV, Greenlees IA, et al. Stressors in elite sport: A coach perspective. J Sports Sci 2008;26(9):905–18.
18. Rhind DJ, Scott M, Fletcher D. Organizational stress in professional soccer coaches. Int J Sport Psychol 2013;44(1):1–16.
19. Srem-Sai M, Hagan JE, Ogum PN, et al. Assessing the prevalence, sources and selective antecedents of organizational stressors among elite football players and coaches in the Ghana premier league: Empirical evidence for applied practice. Frontiers in Sports and Active Living 2022;4:938619.
20. Olusoga P, Butt J, Hays K, et al. Stress in Elite Sports Coaching: Identifying Stressors. J Appl Sport Psychol 2009;21(4):442–59.
21. Baldock L, Cropley B, Neil R, et al. Stress and Mental Well-Being Experiences of Professional Football Coaches. Sport Psychol 2021;35(2):108–22.
22. Kegelaers J, Wylleman P, van Bree INA, et al. Mental Health in Elite-Level Coaches: Prevalence Rates and Associated Impact of Coach Stressors and Psychological Resilience. International Sport Coaching Journal 2021;8(3):338–47.

Downloaded for Anonymous User (n/a) at The Ohio State University from ClinicalKey.com by Elsevier on October 20, 2023. For personal use only. No other uses without permission. Copyright ©2023. Elsevier Inc. All rights reserved.

23. Pilkington V, Rice SM, Walton CC, et al. Prevalence and correlates of mental health symptoms and well-being among elite sport coaches and high-performance support staff. Sports Medicine-Open 2022;8(1):89.
24. Fletcher D, Scott M. Psychological stress in sports coaches: A review of concepts, research, and practice. J Sports Sci 2010;28(2):127–37.
25. Norris LA, Didymus FF, Kaiseler M. Stressors, coping, and well-being among sports coaches: A systematic review. Psychol Sport Exerc 2017;33:93–112.
26. Potts AJ, Didymus FF, Kaiseler M. Psychological stress and psychological well-being among sports coaches: a meta-synthesis of the qualitative research evidence. Int Rev Sport Exerc Psychol 2021;1–30.
27. Olsen MG, Haugan JA, Hrozanova M, et al. Coping Amongst Elite-Level Sports Coaches: A Systematic Review. International Sport Coaching Journal 2021; 8(1):34–47.
28. Olusoga P, Bentzen M, Kentta G. Coach Burnout: A Scoping Review. International Sport Coaching Journal 2019;6(1):42–62.
29. Walton CC, Gwyther K, Gao CX, et al. Evidence of gender imbalance across samples in sport and exercise psychology. Int Rev Sport Exerc Psychol 2022;1–19.
30. Serpell BG, Harrison D, Dower R, et al. The under representation of women coaches in high-performance sport. Int J Sports Sci Coach 2023;0(0):1–13.
31. Kim SSY, Hamiliton B, Beable S, et al. Elite coaches have a similar prevalence of depressive symptoms to the general population and lower rates than elite athletes. BMJ open sport & exercise medicine 2020;6(1):e000719.
32. Åkesdotter C, Kenttä G, Eloranta S, et al. Prevalence and comorbidity of psychiatric disorders among treatment-seeking elite athletes and high-performance coaches. BMJ Open Sport & Exercise Medicine 2022;8(1):e001264.
33. Lee YH. The Roles of Different Appraisals in Anxiety and Emotional Exhaustion: A Case of NCAA Division I Head Coaches. Am J Psychol 2021;134(3):269–83.
34. Ruddock S, Ruddock-Hudson M, Rahimi-Golkhandan S. The impact of job-burnout on Australian Football League coaches: Mental health and well-being. J Sci Med Sport 2017;20:S73–4.
35. Ruddock S, Ruddock-Hudson M, Rahimi-Golkhandan S. Examining within-season change of job-burnout and psychological distress for Australian Rules Football coaches. J Sci Med Sport 2018;21:S41–2.
36. Ruddock S, Rahimi-Golkhanden S, Ruddock-Hudson M, et al. Tracking the mental health outcomes of occupational burnout with Australian Rules Football coaches: A 2-year longitudinal study. J Sci Med Sport 2019;22:S52.
37. Smith A, Torres-McGehee T, Monsma E, et al. Prevalence of Eating Disorder Risk and Body Image Perceptions of Collegiate Cheerleading Coaches. Journal of Sports Medicine & Allied Health Sciences: Official Journal of the Ohio Athletic Trainers' Association 2018;4(1):1–2.
38. Gwyther K, Rice S, Purcell R, et al. Sleep interventions for performance, mood and sleep outcomes in athletes: A systematic review and meta-analysis. Psychol Sport Exerc 2022;58:102094.
39. Bentzen M, Lemyre P-N, Kenttä G. The process of burnout among professional sport coaches through the lens of self-determination theory: a qualitative approach. Sports Coaching Review 2014;3(2):101–16.
40. Lundkvist E, Gustafsson H, Hjälm S, et al. An interpretative phenomenological analysis of burnout and recovery in elite soccer coaches. Qualitiative Research in Sport, Exercise and Health 2012;4(3):400–19.
41. Olusoga P, Kenttä G. Desperate to Quit: A Narrative Analysis of Burnout and Recovery in High-Performance Sports Coaching. Sport Psychol 2017;31(3):237–48.

Downloaded for Anonymous User (n/a) at The Ohio State University from ClinicalKey.com by Elsevier on October 20, 2023. For personal use only. No other uses without permission. Copyright ©2023. Elsevier Inc. All rights reserved.

42. Baldock L, Cropley B, Mellalieu SD, et al. A Longitudinal Examination of Stress and Mental Ill-/Well-Being in Elite Football Coaches. Sport Psychol 2022;36(3): 171–82.
43. Bentzen M, Lemyre P-N, Kenttä G. Development of exhaustion for high-performance coaches in association with workload and motivation: A person-centered approach. Psychol Sport Exerc 2016;22:10–9.
44. Kaski SS, Kinnunen U. Work-related ill- and well-being among Finnish sport coaches: Exploring the relationships between job demands, job resources, burnout and work engagement. International Journal of Sport Science & Coaching 2021;16(2):262–71.
45. Lundkvist E, Gustafsson H, Madigan D, et al. The Prevalence of Emotional Exhaustion in Professional and Semiprofessional Coaches. J Clin Sport Psychol 2022;12:1–14.
46. Nikolaos A. An examination of a burnout model in basketball coaches. Journal of Physical Education & Sport 2012;12(2):171–9.
47. Ryska TA. Multivariate Analysis of Program Goals, Leadership Style, and Occupational Burnout Among Intercollegiate Sport Coaches. J Sport Behav 2009; 32(4):476–88.
48. Gencay S, Gencay OA. Burnout among Judo Coaches in Turkey. J Occup Health 2011;53(5):365–70.
49. Hjälm S, Kenttä G, Hassmén P, et al. Burnout among elite soccer coaches. J Sport Behav 2007;30(4):415–27.
50. Gouttebarge V, Bindra A, Blauwet C, et al. International Olympic Committee (IOC) Sport Mental Health Assessment Tool 1 (SMHAT-1) and Sport Mental Health Recognition Tool 1 (SMHRT-1): towards better support of athletes' mental health. Br J Sports Med 2021;55(1):30–7.
51. Hill DM, Brown G, Lambert T-L, et al. Factors perceived to affect the wellbeing and mental health of coaches and practitioners working within elite sport. Sport, Exercise, and Performance Psychology 2021;10(4):504–18.
52. Walton CC, Rice S, Gao CX, et al. Gender differences in mental health symptoms and risk factors in Australian elite athletes. BMJ open sport & exercise medicine 2021;7(1):e000984.
53. Biggins M, Purtill H, Fowler P, et al. Sleep in elite multi-sport athletes: Implications for athlete health and wellbeing. Phys Ther Sport 2019;39:136–42.
54. Åkesdotter C, Kenttä G, Eloranta S, et al. The prevalence of mental health problems in elite athletes. J Sci Med Sport 2020;23(4):329–35.
55. Hassmén P, Kenttä G, Hjälm S, et al. Burnout symptoms and recovery processes in eight elite soccer coaches over 10 years. Int J Sports Sci Coach 2019;14(4): 431–43.
56. Bentzen M, Lemyre P, Kenttä G. Changes in Motivation and Burnout Indices in High-Performance Coaches Over the Course of a Competitive Season. J Appl Sport Psychol 2016;28(1):28–48.
57. de Sousa Pinheiro G, Túlio de Mello M, Gustavo dos Santos F, et al. Analysis of stress level and recovery of formative football coaches. Case studies. Retos: Nuevas Perspectivas de Educación Física. Deporte y Recreación 2021;41: 345–53.
58. Kellmann M, Altfeld S, Mallett CJ. Recovery–stress imbalance in Australian Football League coaches: A pilot longitudinal study. Int J Sport Exerc Psychol 2016; 14(3):240–9.

Downloaded for Anonymous User (n/a) at The Ohio State University from ClinicalKey.com by Elsevier on October 20, 2023. For personal use only. No other uses without permission. Copyright ©2023. Elsevier Inc. All rights reserved.

ARTICLE IN PRESS

59. Georgios K, Nikolaos A. An Investigation of a Model of Personal-Situational Factors, Stress and Burnout in Track and Field Coaches. Journal of Physical Education & Sport 2012;12(3):343–9.
60. Lee YH, Chelladurai P. Affectivity, Emotional Labor, Emotional Exhaustion, and Emotional Intelligence in Coaching. J Appl Sport Psychol 2016;28(2):170–84.
61. Longshore K, Sachs M. Mindfulness Training for Coaches: A Mixed-Method Exploratory Study 2015;9(2):116–37.
62. Hägglund K, Kenttä G, Thelwell R, et al. Mindful self-reflection to support sustainable high-performance coaching: A process evaluation of a novel method development in elite sport. J Appl Sport Psychol 2021;34:1–24.
63. DeWolfe CEJ, Dithurbide L. Beware of the blues: Wellbeing of coaches and support staff throughout the Olympic Games. Int J Sports Sci Coach 2022;17(6):1243–57.
64. Bentzen M, Kenttä G, Lemyre P-N. Elite Football Coaches Experiences and Sensemaking about Being Fired: An Interpretative Phenomenological Analysis. Int J Environ Res Publ Health 2020;17(14):5196.
65. Kenttä G, Mellalieu S, Roberts C-M. Are Career Termination Concerns Only for Athletes? A Case Study of the Career Termination of an Elite Female Coach. Sport Psychol 2016;30(4):314–26.
66. Ng JYY, Ntoumanis N, Thøgersen-Ntoumani C, et al. Self-Determination Theory Applied to Health Contexts: A Meta-Analysis. Perspect Psychol Sci 2012;7(4):325–40.
67. Mrazek PJ, Haggerty RJ. Reducing risks for mental disorders: Frontiers for preventive intervention research. Washington, DC, US: National Academy Press; 1994.
68. Frost J, Walton CC, Purcell R, et al. Supporting the mental health of elite-level coaches through early intervention. Arthroscopy, Sports Medicine, and Rehabilitation 2023. https://doi.org/10.1016/j.asmr.2023.04.017.
69. Sullivan P, Murphy J, Blacker M. The Level of Mental Health Literacy Among Athletic Staff in Intercollegiate Sport. J Clin Sport Psychol 2019;13(3):440–50.
70. Purcell R, Pilkington V, Carberry S, et al. An Evidence-Informed Framework to Promote Mental Wellbeing in Elite Sport. Front Psychol 2022;13:780359.
71. Rice S, Walton CC, Pilkington V, et al. Psychological safety in elite sport settings: a psychometric study of the Sport Psychological Safety Inventory. BMJ Open Sport and Exercise Medicine 2022;8(2):e001251.
72. Vella SA, Mayland E, Schweickle MJ, et al. Psychological safety in sport: a systematic review and concept analysis. Int Rev Sport Exerc Psychol 2022;1–24.

Downloaded for Anonymous User (n/a) at The Ohio State University from ClinicalKey.com by Elsevier on October 20, 2023. For personal use only. No other uses without permission. Copyright ©2023. Elsevier Inc. All rights reserved.